

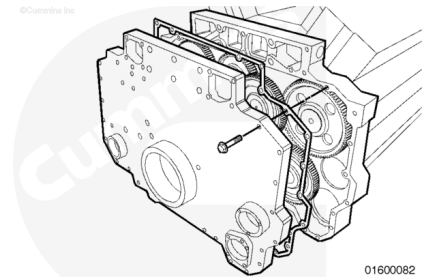
Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

This assembly weighs more than 23 kg [50 lb]. To reduce the possibility of personal injury, use a hoist or get personal assistance to lift this assembly.

- Remove the front gear cover. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/56/56-001-031-tr.html>)



Remove

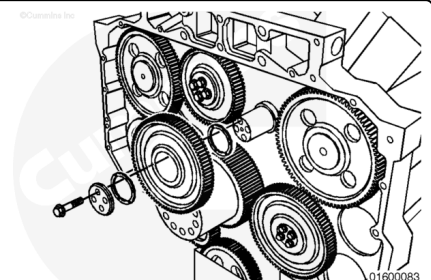
Note : The left bank idler gear is shown in the illustration and **must** be removed first.

Cummins Inc., recommends measuring the end play before removing the idler gear.

Remove the four capscrews, the thrust retainer plate, and the thrust bearings on the left-bank camshaft idler gear.

Remove the camshaft idler gear.

Remove the idler shaft.



Clean and Inspect for Reuse

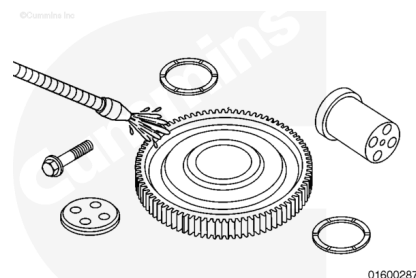
⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the

possibility of personal injury.

⚠ WARNING ⚠

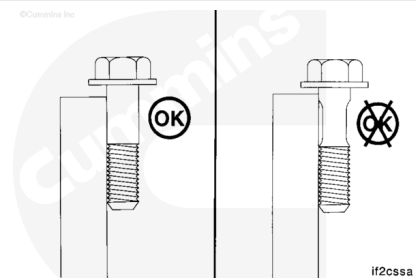
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Clean the parts with solvent and dry with compressed air.

Use a straightedge to inspect the retaining capscrews for necking.

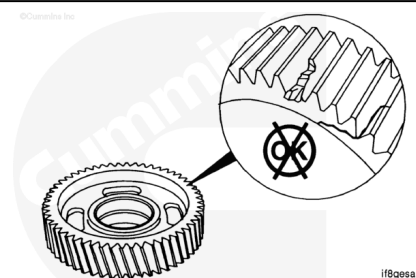
If the retaining capscrews are necked, they **must** be replaced.



Inspect the idler gear for chipped, broken, or cracked teeth.

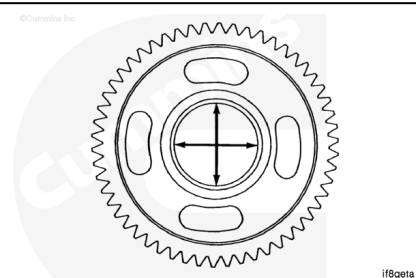
Inspect the bushing for damage.

If the idler gear bushing or teeth are damaged, the idler gear **must** be replaced.



Measure the inside diameter of the idler gear bushing.

Right Bank Camshaft Idler Gear Inside Diameter		
mm		in
73.035	MIN	2.8753
73.065	MAX	2.8765



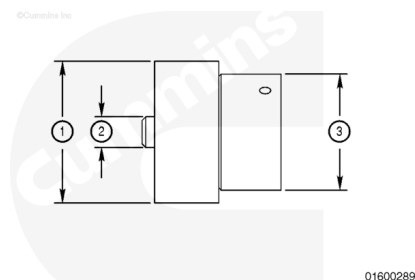
The bushing in the idler gear is precision bored after installation. Replace the bushing and gear as an assembly if the idler gear bushing is **not** within specification.

If the idler gear is a compound gear it **must** be replaced as an assembly.

Measure the outside diameter of the idler gear shaft.

Right Bank Camshaft Idler Gear Shaft Outside Diameter

	mm		in
(1)	88.75	MIN	3.494
	89.05	MAX	3.506
(2)	21.975	MIN	0.8652
	22.000	MAX	0.8661
(3)	72.92	MIN	2.871
	72.98	MAX	2.873



If the idler gear shaft is **not** within specifications, it **must** be replaced.

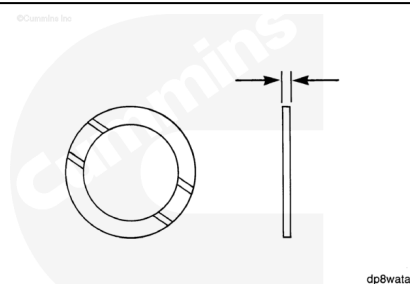
Check the grooved side of the thrust bearing for damage.

If the grooved side of the thrust bearing is damaged, it **must** be replaced.

Measure the thrust bearing thickness.

Right Bank Camshaft Idler Gear Thrust Bearing Thickness

mm		in
2.337	MIN	0.092
2.387	MAX	0.094



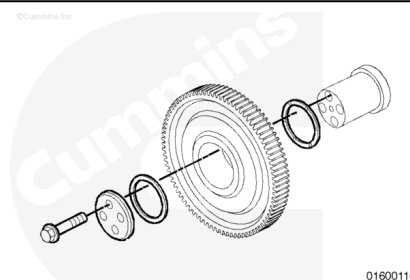
If the thrust bearing is **not** within specification, it **must** be replaced.

Install

The grooves in the thrust washer **must** be positioned toward the gear.

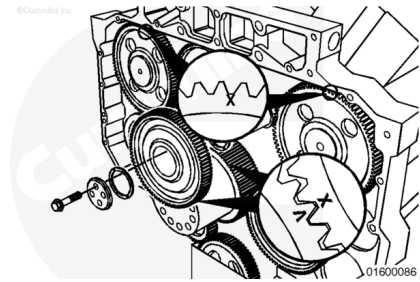
Use Lubriplate™ 105, or equivalent. Lubricate the gear bushing, shaft, and thrust washer.

Lubricate the capscrews with clean engine oil. Assemble the parts as illustrated.



Note : The left bank idler gear is shown in the illustration and **must** be installed last.

Align the “V” on the idler with the “X” on the crankshaft gear.
Align the “X” on the camshaft gears in the 12-o'clock position,
install the camshaft locking pin.

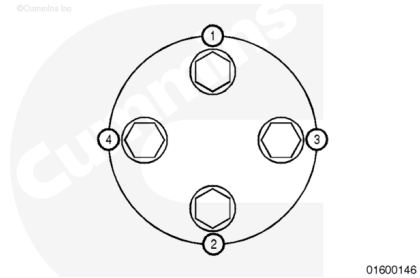


Rotate the idler shaft as necessary to align the mounting
capscrew holes.

The oil holes in the idler shaft **must** be placed in the 11-o'clock
position on the right-bank idler shaft.

Use the capscrews to pull the shaft into the bore.

Use the following steps to tighten the capscrews in the
sequence shown.

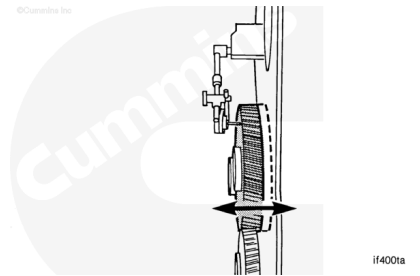


Torque Value:

1. 80 n•m [59 ft-lb]
2. 165 n•m [122 ft-lb]
3. 280 n•m [207 ft-lb]

Use a dial indicator to measure the idler gear end clearance.

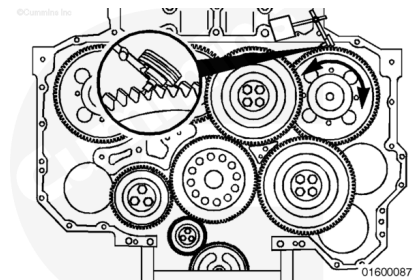
Right Bank Camshaft Idler Gear End Clearance		
mm		in
0.15	MIN	0.006
0.33	MAX	0.013



If the clearance is **not** within specifications, check for foreign
material between the parts and proper location of the thrust
washers.

Measure the front gear train backlash.

Front Gear Train Backlash		
mm		in
0.07	MIN	0.003
0.51	MAX	0.020



Finishing Steps

⚠ WARNING ⚠

This assembly weighs more than 23 kg [50 lb]. To reduce the possibility of personal injury, use a hoist or get personal assistance to lift this assembly.

- Install the front gear cover. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/56/56-001-031-tr.html>)
- Operate the engine and check for leaks.

